WEEK 1:

Module 1 - Key Takeaways

- Understood and applied SOLID principles for building maintainable and scalable software.
- Explored design patterns such as Singleton, Factory Method, and Adapter to solve common design problems.
- Enhanced code reusability, flexibility, and overall software design through practical implementation.

Exercise 1: Implementing the Singleton Pattern

Scenario:

You need to ensure that a logging utility class in your application has only one instance throughout the application lifecycle to ensure consistent logging.

Steps:

1. **Create a New Java Project:**
    - Create a new Java project named **SingletonPatternExample**.
2. **Define a Singleton Class:**
    - Create a class named Logger that has a private static instance of itself.
    - Ensure the constructor of Logger is private.
    - Provide a public static method to get the instance of the Logger class.
3. **Implement the Singleton Pattern:**
    - Write code to ensure that the Logger class follows the Singleton design pattern.
4. **Test the Singleton Implementation:**
    - Create a test class to verify that only one instance of Logger is created

Code :


```using System;

class Logger
{
    private static Logger instance = new Logger();

    private Logger() { }

    public static Logger GetInstance()
    {
        return instance;
    }

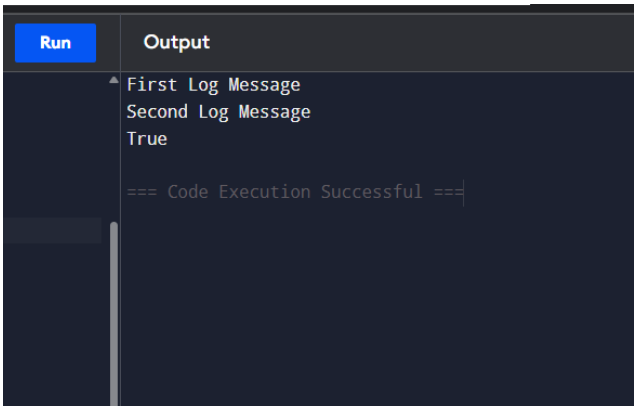
    public void Log(string message)
    {
        Console.WriteLine(message);
    }
}

class Program
{
    static void Main(string[] args)
    {
        Logger logger1 = Logger.GetInstance();
        Logger logger2 = Logger.GetInstance();

        logger1.Log("First Log Message");
        logger2.Log("Second Log Message");

        Console.WriteLine(logger1 == logger2);
    }
}```

# OUTPUT :


```

Exercise 2: Implementing the Factory Method Pattern

Scenario:

You are developing a document management system that needs to create different types of documents (e.g., Word, PDF, Excel). Use the Factory Method Pattern to achieve this.

Steps:

1. **Create a New Java Project:**
    - Create a new Java project named **FactoryMethodPatternExample**.
2. **Define Document Classes:**
    - Create interfaces or abstract classes for different document types such as **WordDocument**, **PdfDocument**, and **ExcelDocument**.
3. **Create Concrete Document Classes:**
    - Implement concrete classes for each document type that implements or extends the above interfaces or abstract classes.
4. **Implement the Factory Method:**
    - Create an abstract class **DocumentFactory** with a method **createDocument()**.
    - Create concrete factory classes for each document type that extends DocumentFactory and implements the **createDocument()** method.
5. **Test the Factory Method Implementation:**

    Create a test class to demonstrate the creation of different document types using the factory method.

Code :

Main.cs

```
1  using System;
2  interface IDocument
3  {
4      void Open();
5  }
6  class WordDocument : IDocument
7  {
8      public void Open()
9      {
10         Console.WriteLine("Word Document Opened");
11     }
12 }
13 class PdfDocument : IDocument
14 {
15     public void Open()
16     {
17         Console.WriteLine("PDF Document Opened");
18     }
19 }
20 class ExcelDocument : IDocument
21 {
22     public void Open()
23     {
24         Console.WriteLine("Excel Document Opened");
25     }
```

Output :

Output

```
Word Document Opened
PDF Document Opened
Excel Document Opened

=== Code Execution Successful ===
```